

NMAP Full Command

-- Architect Johan

1. TARGET SPECIFICATION

Command :

nmap 192.168.1.1

👉 Ek IP scan

👉 Kyun: Kisi ek device ka pura port status check karne ke liye

👉 Example: Apne WiFi router ka scan — pata chalega admin port (80/443) open hai ya nahi

👉 Extra: Agar tu 192.168.1.1 ko scan karega to tere ISP ka modem dikhega

Command :

nmap scanme.nmap.org

👉 Domain scan

👉 Kyun: Bina IP jaane directly website target karne ke liye

👉 Example: Nmap khud ka demo website deta hai legal practice ke liye

👉 Extra: Real bug bounty me domain name se start karte hain

Command :

nmap 192.168.1.0/24

👉 Poora network scan (256 IPs)

👉 Kyun: Pata karna ki network mein kaun-kaun connected hai

👉 Example: Office mein 50 computers hain — sab ek saath scan

👉 Extra: `/24` ka matlab 255.255.255.0 subnet mask

Command :

nmap 10.0.0-255.1-254

👉 Custom IP range

👉 Kyun: Specific portion scan karna, pure /16 nahi

👉 Example: 10.0.0.x se 10.0.255.x ke beech ki IPs

👉 Extra: Ye format `nmap` unique hai — bahut powerful

Command :

nmap -iL targets.txt

👉 File se IP list scan

- 👉 Kyun: Jab 100+ targets ho, ek-ek type karna mushkil
- 👉 Example: targets.txt mein 100 bug bounty domains
- 👉 Extra: Har line mein ek IP ya domain

Command :

nmap -iR 10

- 👉 Random 10 IPs scan
- 👉 Kyun: Internet random discovery — sirf research
- 👉 Example: Kisi unknown server pe kya chal raha hai
- 👉 Extra: ⚠️ Illegal hai bina permission ke. Sirf apne lab mein

Command :

nmap 192.168.1.0/24 --exclude 192.168.1.5

- 👉 Ek IP skip
- 👉 Kyun: Apna khud ka system scan mein nahi aana chahiye
- 👉 Example: Tera IP 192.168.1.5 hai to usko exclude kar
- 👉 Extra: Multiple IPs comma se likh sakta hai: `--exclude 1.1,1.2`

Command :

nmap 192.168.1.0/24 --excludefile exclude.txt

- 👉 File se multiple IP skip
- 👉 Kyun: 10-20 sensitive servers hain jo nahi scan karne
- 👉 Example: exclude.txt mein production server IPs
- 👉 Extra: Production server kabhi mat scan karna bina approval ke

❑ 2. HOST DISCOVERY

Command :

nmap -sL 192.168.1.0/24

- 👉 Sirf list, scan nahi
- 👉 Kyun: Planning ke liye — kaunse IPs exist karte hain
- 👉 Example: Network diagram banana hai
- 👉 Extra: Koi packet nahi bhejta, DNS se resolve karta hai

Command :

nmap -sn 192.168.1.0/24

- 👉 Kaun alive hai (ping only)

- 👉 Kyun: Port scan se pehle pata karna konse systems on hain
- 👉 Example: 200 IPs mein sirf 30 live hain — unhi ko scan karo
- 👉 Extra: Isko "ping sweep" bhi bolte hain

Command :

nmap -Pn 192.168.1.1

- 👉 Ping skip karke direct scan
- 👉 Kyun: Firewall ICMP block karta hai to bhi scan chahiye
- 👉 Example: AWS security group ne ping drop kiya — still scan
- 👉 Extra: Slow hota hai, sab ports try karega

TCP Discovery

Command :

nmap -PS80 192.168.1.1

- 👉 SYN ping on port 80
- 👉 Kyun: ICMP block ho to TCP SYN se check karo alive hai ya nahi
- 👉 Example: Web server ka port 80 open hai to alive
- 👉 Extra: Multiple ports: `-PS22,80,443`

Command :

nmap -PA80 192.168.1.1

- 👉 ACK ping
- 👉 Kyun: Firewall rules ko test karna
- 👉 Example: Stateful firewall ignore karta hai ACK packets

Command :

nmap -PU53 192.168.1.1

- 👉 UDP ping (DNS port)
- 👉 Kyun: UDP service check karna
- 👉 Example: DNS server alive hai ya nahi

Command :

nmap -PY80 192.168.1.1

- 👉 SCTP ping
- 👉 Kyun: Telecom / SS7 networks mein use hota hai
- 👉 Example: VoIP server discovery

ICMP

Command :

nmap -PE 192.168.1.1

- ✍ Normal ICMP echo ping
- ✍ Kyun: Sabse standard alive check
- ✍ Example: `ping` command jaisa but nmap style

Command :

nmap -PP 192.168.1.1

- ✍ ICMP timestamp request
- ✍ Kyun: Kuch firewalls echo block karte hain lekin timestamp allow
- ✍ Example: Bypass technique

Command :

nmap -PM 192.168.1.1

- ✍ Netmask request
- ✍ Kyun: Rare, lekin kabhi kaam aata hai
- ✍ Example: Old network devices

Command :

nmap -PO1 192.168.1.1

- ✍ Protocol ping (IP protocol 1 = ICMP)
- ✍ Kyun: Layer 3 protocol scan
- ✍ Example: Protocol number 6 (TCP) ya 17 (UDP) bhi de sakta hai

DNS

Command :

nmap -n 192.168.1.1

- ✍ DNS resolution band
- ✍ Kyun: Fast scan — DNS query ka time bachta hai
- ✍ Example: Internal network scan

Command :

nmap -R 192.168.1.1

- ✍ Hamesha DNS resolve
- ✍ Kyun: Domain names chahiye result mein
- ✍ Example: Report banani hai to IP ke saath name bhi chahiye

Command :

nmap --dns-servers 8.8.8.8 192.168.1.1

- ✍ Custom DNS server use
- ✍ Kyun: Default DNS slow ya unreliable ho

👉 Example: Google DNS ya Cloudflare (1.1.1.1) use karna

Command :

nmap --system-dns 192.168.1.1

👉 OS ka DNS use karo

👉 Kyun: `/etc/resolv.conf` ya Windows registry wala DNS

👉 Example: Corporate DNS ke through resolve karna

Command :

nmap --traceroute 192.168.1.1

👉 Network path find karo

👉 Kyun: Kitne routers ke paar target hai

👉 Example: Local se Google tak ka route

👉 Extra: `traceroute` command jaisa but nmap style



3. SCAN TECHNIQUES

Command :

nmap -sS 192.168.1.1

👉 Stealth SYN scan (half-open)

👉 Kyun: Full TCP connection nahi banata — log files mein kam dikhta

👉 Example: Hacker ka favorite — fast aur quiet

👉 Extra: Root/Admin chahiye iske liye

Command :

nmap -sT 192.168.1.1

👉 Full connect scan

👉 Kyun: Jab raw packets nahi bhej sakte (no root)

👉 Example: Windows machine se scan

👉 Extra: Detectable hai — `connect()` system call use karta hai

Command :

nmap -sA 192.168.1.1

👉 ACK scan — firewall rule detection

👉 Kyun: Pata karna packet filtered hai ya nahi

👉 Example: Port open nahi dikhega, sirf "filtered" vs "unfiltered"

Command :

nmap -sW 192.168.1.1

👉 Window scan

👉 Kyun: Old TCP stack ki vulnerability use karna

👉 Example: Kuch systems mein ACK scan se better hai

Command :

`nmap -sM 192.168.1.1`

- 📖 Maimon scan (FIN/ACK)
- 📖 Kyun: Rare — kuch IDS confuse karne ke liye
- 📖 Example: RFC 793 ke hisaab se ye invalid packet hai

Command :

`nmap -sU 192.168.1.1`

- 📖 UDP scan
- 📖 Kyun: DNS, SNMP, DHCP jaise UDP services detect
- 📖 Example: DNS server ka port 53 open hai?
- 📖 Extra: 🧠 Slow — ICMP port unreachable ka wait karta hai

Command :

`nmap -sN 192.168.1.1` NULL scan

`nmap -sF 192.168.1.1` FIN scan

`nmap -sX 192.168.1.1` XMAS scan

- 📖 Stealth weird scans
- 📖 Kyun: Non-TCP standard packets — kuch firewalls ignore karte hain
- 📖 Example: XMAS scan mein FIN, PSH, URG flags ON
- 📖 Extra: Windows systems pe kaam nahi karte ye

Command :

`nmap --scanflags SYNFIN 192.168.1.1`

- 📖 Custom TCP flags set
- 📖 Kyun: Fully custom packet bhejna hai
- 📖 Example: SYN+RST flag combination — very weird
- 📖 Extra: `--scanflags RSTSYNFIN` bhi likh sakta hai

Command :

`nmap -sI zombie_ip 192.168.1.1`

- 📖 Idle scan (anonymous)
- 📖 Kyun: Tera real IP kahin log nahi hoga
- 📖 Example: Zombie machine (inactive) ke through scan
- 📖 Extra: 🐱 Truly anonymous — but zombie chahiye

Command :

`nmap -sY 192.168.1.1 SCTP INIT`

`nmap -sZ 192.168.1.1 SCTP COOKIE-ECHO`

☞ SCTP scan

☞ Kyun: Telecom, SS7, SIGTRAN networks

☞ Example: VoIP server discovery

Command :

`nmap -sO 192.168.1.1`

☞ IP protocol scan

☞ Kyun: Kaunse Layer-3 protocols supported hain

☞ Example: TCP (6), UDP (17), ICMP (1), GRE (47)

Command :

`nmap -b ftp.example.com 192.168.1.1`

☞ FTP bounce scan

☞ Kyun: FTP server ke through scan — source IP chhupana

☞ Example: Vulnerable FTP server proxy ki tarah use karna

☞ Extra: Bohot purana technique — ab rarely kaam karta hai

4. PORT OPTIONS

Command :

`nmap -p 22 192.168.1.1`

☞ Sirf ek port

☞ Kyun: SSH check karna hai, baaki nahi

☞ Example: SSH bruteforce se pehle port 22 open hai ya nahi

Command :

`nmap -p 1-65535 192.168.1.1`

☞ Full port scan (65k ports)

☞ Kyun: Koi port miss nahi hona chahiye

☞ Example: Pentest full coverage

☞ Extra: Time lagta hai — hours bhi ho sakta hai

Command :

`nmap -p U:53,111,T:80,443 192.168.1.1`

- 👉 UDP aur TCP mix scan
- 👉 Kyun: Ek command mein dono protocols
- 👉 Example: DNS (UDP53) + HTTP/HTTPS (TCP80/443)

Command :

`nmap --exclude-ports 22,80 192.168.1.1`

- 👉 Kuch ports skip karo
- 👉 Kyun: Known services ko scan nahi karna
- 👉 Example: SSH aur HTTP already safe hai — baaki check karo

Command :

`nmap -F 192.168.1.1`

- 👉 Fast scan (100 common ports)
- 👉 Kyun: Time kam hai — jaldi result chahiye
- 👉 Example: Quick reconnaissance
- 👉 Extra: Default se 10x faster

Command :

`nmap -r 192.168.1.1`

- 👉 Sequential port scan
- 👉 Kyun: Randomize nahi karna port order
- 👉 Example: Debugging — predictable order

Command :

`nmap --top-ports 100 192.168.1.1`

- 👉 Top 100 common ports
- 👉 Kyun: Statistics ke hisaab se sabse zyada open ports
- 👉 Example: Nmap ki internal frequency list use karta hai

Command :

`nmap --port-ratio 0.5 192.168.1.1`

- 👉 Frequent ports (50%+ open in database)
 - 👉 Kyun: Sirf popular ports scan karna
 - 👉 Example: Nmap's `nmap-services` file ke ratio ke hisaab se
-

🦴 5. VERSION DETECTION

Command :

nmap -sV 192.168.1.1

- 👉 Service aur version pata karo
- 👉 Kyun: Sirf port open nahi — kaunsi software chal rahi hai
- 👉 Example: Apache 2.4.41 ya nginx 1.18.0
- 👉 Extra: Vulnerability matching ke liye zaroori hai

Command :

nmap --version-intensity 5 192.168.1.1

- 👉 Probe intensity 0-9 (5 medium)
- 👉 Kyun: Balance speed vs accuracy
- 👉 Example: Default 7 hai — 5 kam aggressive

Command :

nmap --version-light 192.168.1.1

- 👉 Fast version scan (intensity 2)
- 👉 Kyun: Jaldi result — sirf obvious probes
- 👉 Example: Time kam hai to ye use karo

Command :

nmap --version-all 192.168.1.1

- 👉 Full aggressive version scan (intensity 9)
- 👉 Kyun: Har possible probe bhejega — most accurate
- 👉 Example: Deep enumeration — lekin slow 🦴

Command :

nmap --version-trace 192.168.1.1

- 👉 Debug version detection
- 👉 Kyun: Kaunsa probe kis port pe bheja — debugging
- 👉 Example: Nmap developers use karte hain

🦴 6. SCRIPT SCAN

Command :

nmap -sC 192.168.1.1

- 👉 Default script scan
- 👉 Kyun: Basic enumeration — safe aur useful scripts
- 👉 Example: SSH host key, HTTP title, SMB shares
- 👉 Extra: `--script=default` jaisa hi hai

Command :

```
nmap --script=vuln 192.168.1.1
```

📁 Vulnerability detection scripts

📁 Kyun: Known CVEs ka check

📁 Example: EternalBlue (MS17-010), Heartbleed

📁 Extra: Sirf check karta hai — exploit nahi karta

Command :

```
nmap --script=http-enum 192.168.1.1
```

📁 Web directory enumeration

📁 Kyun: Hidden directories/files dhundo

📁 Example: `/admin`, `/backup.zip`, `/phpinfo.php`

📁 Extra: Dirb/gobuster jaisa but nmap style

Command :

```
nmap --script "default,safe" 192.168.1.1
```

📁 Sirf safe scripts

📁 Kyun: Crash ya DoS ka risk nahi lena

📁 Example: Production environment scan

Command :

```
nmap --script-args user=admin,pass=1234 192.168.1.1
```

📁 Script ko arguments dena

📁 Example: HTTP basic auth brute force ke liye user/pass

📁 Extra: Multiple args comma se separate

Command :

```
nmap --script-args-file args.txt 192.168.1.1
```

📁 File se arguments

📁 Kyun: Bohot saare arguments ho to

📁 Example: args.txt mein `user=admin\npass=1234`

Command :

```
nmap --script-trace 192.168.1.1
```

📁 Har script packet ka debug output

📁 Kyun: Script kaise kaam kar rahi hai — analysis

📁 Example: Custom script debugging

Command :

```
nmap --script-updatedb
```

📁 Nmap script database update

📁 Kyun: Naye scripts download karna

📁 Example: `nmap --script-updatedb` run karo monthly

Command :

nmap --script-help vuln

- 👉 Script ke baare mein info
 - 👉 Kyun: Script kya karti hai — documentation
 - 👉 Example: `--script-help http-enum` -> usage + description
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📦 7. OS DETECTION

Command :

nmap -O 192.168.1.1

- 👉 OS fingerprinting
- 👉 Kyun: Target Windows/Linux/Mac/FreeBSD?
- 👉 Example: "Linux 5.4", "Windows 10", "Cisco IOS"
- 👉 Extra: Root/Admin chahiye — raw packets bhejta hai

Command :

nmap --osscan-limit 192.168.1.1

- 👉 Sirf promising targets pe OS scan
- 👉 Kyun: Time bachao — jab sure ho ki port open hai tabhi OS detect
- 👉 Example: `-O` ke saath use karo

Command :

nmap --osscan-guess 192.168.1.1

- 👉 Aggressive OS guess
 - 👉 Kyun: Exact match na ho to closest guess dega
 - 👉 Example: Jab confident hona hai to ye use karo
-

⚡ 8. TIMING

Command :

nmap -T0 192.168.1.1

nmap -T5 192.168.1.1

- 👉 T0 = Paranoid (very slow), T5 = Insane (very fast)
- 👉 Kyun: IDS evasion (T0) ya speed (T5)
- 👉 Example: T0: 5 minutes per port — almost undetectable. T5: 1 second full scan — but loud
- 👉 Extra: Default T3 (Normal)

Baki timing commands summary:

Option	Meaning
--min-hostgroup, --max-hostgroup	Parallel hosts (ek time pe kitne hosts scan honge)
--min-parallelism, --max-parallelism	Parallel probes (ek host pe ek time me kitne packets bhejne)
--min-rtt-timeout, --max-rtt-timeout	Wait time for response (packet ka reply kitna wait kare)
--host-timeout	Overall timeout per host (ek host pe max kitna time lagana)
--scan-delay, --max-scan-delay	Delay between probes (har packet ke beech gap)
--min-rate, --max-rate	Packets per second (kitni speed se packets bhejne)

🔒 9. FIREWALL BYPASS

Command :

nmap -f 192.168.1.1

👉 Packet fragmentation

👉 Kyun: 8-byte fragments mein tod ke bhej — kuch firewalls reassemble nahi karte

👉 Example: ``-f`` = 8 bytes, ``-ff`` = 16 bytes

Command :

nmap -D RND:5 192.168.1.1

👉 Decoy IPs — fake sources

👉 Kyun: Real IP chhupana — IDS confuse

👉 Example: 5 random fake IPs ke saath tera real IP mix ho jayega

👉 Extra: ``-D decoy1,decoy2,ME,decoy3``

Command :

nmap -S 192.168.1.100 192.168.1.1

👉 Source IP spoof

👉 Kyun: Bina response chahiye (like idle scan)

👉 Example: ``-S 1.2.3.4 -e eth0``

👉 Extra: Response nahi milega unless tu network path mein ho

Command :

nmap --spoof-mac 00:11:22:33:44:55 192.168.1.1

👉 Fake MAC address

👉 Kyun: Network layer tracking avoid

👉 Example: ``--spoof-mac 0`` = random vendor, ``--spoof-mac Apple`` = Apple MAC

👉 Extra: MAC spoofing sirf local network mein kaam karta hai

Baki bypass techniques:

Option	Meaning
--data-length 200	Extra random data add karo
--ttl 64	TTL value change
--randomize-hosts	Host order random
--mtu 24	Custom MTU for fragmentation

10. OUTPUT

Command :

nmap -oN result.txt 192.168.1.1

- 👉 Normal output save
- 👉 Kyun: Human readable — exactly terminal jaisa
- 👉 Example: Report ke liye use karo

Command :

nmap -oX result.xml 192.168.1.1

- 👉 XML output
- 👉 Kyun: Parsing ke liye — tools ke liye
- 👉 Example: Metasploit import karta hai XML

Command :

nmap -oG result.gnmap 192.168.1.1

- 👉 Grepable output
- 👉 Kyun: `grep` aur scripts ke liye easy
- 👉 Example: `cat result.gnmap | grep "open"``

Command :

nmap -oA fullscan 192.168.1.1

- 👉 All formats ek saath
- 👉 Kyun: Normal + XML + Grepable dono chahiye
- 👉 Example: `fullscan.nmap`, `fullscan.xml`, `fullscan.gnmap` banenge

11. MISC (Miscellaneous)

Command :

nmap -A 192.168.1.1

- 👉 Aggressive scan — sab kuch

- 👉 Kyun: OS detection + Version + Scripts + Traceroute ek command mein
- 👉 Example: Pentest start — `nmap -A -T4 192.168.1.0/24`
- 👉 Extra: Loud hai — logs mein clearly dikhega

Command :

nmap -V

- 👉 Version check
- 👉 Kyun: Konsa version installed hai
- 👉 Example: `nmap -V | grep "Nmap version"`

Command :

nmap -h

- 👉 Help menu
- 👉 Kyun: Command yaad nahi aa raha
- 👉 Example: `nmap -h | grep "scan"`

🦴 FINAL UNDERSTANDING

- 👉 Nmap ka flow (exact 6 steps):

Step	Description
1. Target choose	IP, range, file, random
2. Host discovery	Kaun alive hai (ping/ARP/TCP/UDP)
3. Port scan	Kaunse ports open/closed/filtered
4. Service/Version detect	Kaunsi software + version
5. OS detection	Windows/Linux/Cisco/Firewall
6. Script scan	Vulnerabilities + enumeration

🦴 FUN MEMORY

- 👉 "Nmap = Shaadi.com nahi... CID investigation tool hai"

Reason:

- Shaadi.com → matchmaking (relationship)
- CID investigation → har suspect ki poori profile (ports, services, OS, vuln)

REAL HACKING SCENARIO (Bonus)

Maan le tu apne friend ke local server ko test kar raha hai (permission se):

Command :

Step 1: Alive hosts find

`nmap -sn 192.168.1.0/24`

Step 2: Quick scan on live host

`nmap -F 192.168.1.100`

Step 3: Deep scan on interesting ports

`nmap -sS -sV -O -p 22,80,443,3306 192.168.1.100`

Step 4: Vulnerability scan

`nmap --script=vuln -p 80,443 192.168.1.100`

Step 5: Full aggressive (if safe)

`nmap -A -T4 192.168.1.100 -oA final_report`

----- THE END -----

 Architect Johan – Official Links

 **Instagram**

 https://www.instagram.com/architect_johan

 **YouTube**

 https://youtube.com/@architect_johan

 **Telegram**

 https://t.me/Architect_Johan

 **Website**

 <https://architect-johan-secure.onrender.com>